RESEARCH & ANALYTICS REPORT | SQL & Tableau Workflow

Student Academic Performance Analysis

An investigation into habits, lifestyle choices, and demographics associated with exam outcomes using a dataset of 1,000 students.

PREPARED FOR

Academic Analytics & Institutional Research

Data scope: N = 1,000 Student Records

ANALYSIS DATE

June 2026

Version 1.0 (Final)

WORKFLOW STACK

PostgreSQL | Tableau Desktop

01. Project Overview & Scope

SECTION I

The objective of this project was to identify the key factors that influence student academic performance and determine which habits, lifestyle choices, and demographic characteristics are associated with higher exam scores.

By analyzing the lifestyle choices and demographic parameters of a large, representative student population, institutions can implement targeted support strategies. This project followed a complete analytics workflow involving data extraction and segmentation through SQL, followed by exploratory visualization and dashboard creation in Tableau.

WORKFLOW SUMMARY

1. **Data Preparation:** Processed a database containing 1,000 student records.
2. **SQL Queries:** Extracted specific cohorts to examine factors such as sleep, screen time, and parental education.
3. **Tableau Visualization:** Built interactive dashboards showing trends and relationships.

02. Dataset Description

The dataset covers 1,000 students across 15 distinct variables, structured into five major categories:

DM

DEMOGRAPHIC

- Age
- Gender

Individual profile metrics

AC

ACADEMIC

- Study Hours / Day
- Attendance %
- Exam Score

Primary performance metrics

LF

LIFESTYLE

- Sleep Hours
- Diet Quality
- Mental Health
- Exercise Frequency

Wellness and habit metrics

ENT

ENTERTAINMENT

- Social Media Hours
- Netflix Hours

Screen-time metrics

EXT

EXTERNAL

- Part-Time Job
- Extracurriculars
- Parental Education
- Internet Quality

Environmental factors

03. Data Extraction & SQL Analysis

SECTION II

Before dashboard construction, SQL queries isolated target student cohorts to analyze how specific lifestyle, demographic, and educational parameters correlated with exam scores. This filtering established baseline comparisons for the final Tableau dashboard.

01 **Parental Education Analysis**
Aggregated and calculated average exam scores across categories of parental education level to see if background impacts performance.

02 **Lifestyle & Wellness Querying**
Filtered student populations on overlapping wellness metrics: diet quality, mental health rating, and sleep duration to study compound effects of physical and mental health.

03 **Entertainment Consumption Analysis**
Segmented students by high vs. low screen-time using combined metrics (social media + Netflix hours) to identify thresholds of academic distraction.

04 **Part-Time Employment Split**
Created distinct datasets comparing employed vs. unemployed students to evaluate time constraints and academic trade-offs.

05 **Attendance Stratification**
Grouped students into attendance percentage tiers to measure the direct relationship between physical class presence and exam marks.

04. Dashboard Development in Tableau

Following SQL prep, the datasets were exported to Tableau. Dashboards were structured around core comparisons, prioritizing intuitive visualization layouts. Designs emphasized highlighting academic drivers and enabling quick cross-group cohort filtering.

04. Key Findings & Insights

SECTION III

01. CLASS PARTICIPATION

1.05x Impact

Attendance is a Dominant Performance Predictor

Comparison between high and low attendance cohorts reveals clear differences. Students with higher attendance consistently achieve examination results of **about 1.05x** as compared to those with low attendance, highlighting the critical nature of classroom participation.

02. HABITS

Strong Trend

Study Hours Directly Correlate with High Scores

Line charts comparing study hours and exam scores display a strong upward trend. Invested study time translates directly into higher marks. Notably, this positive correlation holds true regardless of the student's gender.

03. DISTRACTIONS

Negative Drag

Excessive Online Entertainment Decreases Scores

Students categorized as "chronically online" (high social media and streaming consumption) scored noticeably lower than peers with moderate digital consumption. Tableau dashboards suggest excessive media consumption directly impacts available study time.

04. WELL-BEING

1.20x Impact

Mental Health & Lifestyle Habits Matter Greatly

SQL analyses revealed that healthy habits (better sleep, diet quality, and higher mental health ratings) correlate with stronger outcomes. Students with highly rated well-being scored **an average of 1.2x** higher than those with poor ratings.

05. BACKGROUND

Limited Impact

Parental Education Has Minimal Influence

A treemap comparing parental education levels showed only minor differences in student exam performance. Individual habits and lifestyle choices play a significantly larger role than parental background in this dataset.

06. EMPLOYMENT

Negligible Effect

Part-Time Jobs Do Not Impede Academic Success

Contrary to the common assumption that part-time jobs reduce academic performance, SQL queries and dashboards show only small, negligible score differences between employed and unemployed students.

05. Visualization Analysis

SECTION IV

✓ MOST EFFECTIVE VISUALIZATIONS

Attendance vs. Exam Score Analysis
Demonstrated one of the strongest relationships in the dataset and provided clear, actionable insights for policy makers.

Study Hours Trend Charts
Line charts showed performance increases with study time. Importantly, it highlighted that high-attendance students score higher than low-attendance peers when study hours are identical.

Chronically Online vs. Non-Chronically Online
Side-by-side cohort comparisons showed low-screen-time students have a clear academic edge over high-screen-time students.

Parental Education Treemap
Visually engaging layout summarizing scores, demonstrating that parental background has minimal bearing on academic success.

✗ VISUALIZATIONS WITH LIMITED INSIGHT

Age Bubble Chart
Although visually appealing, the bubble chart showed only minor variations in average scores across age groups, failing to yield actionable insights. A simpler visualization would suffice.

Extracurricular Participation Chart
Negligible performance differences were found between active and non-active students. Extracurriculars do not hurt academics and should be viewed as opportunities for personal/social growth.

06. Strategic Conclusion

The analysis demonstrates that academic performance is primarily driven by classroom attendance and study habits. Conversely, static demographic traits (parental education, gender, age) and part-time jobs show negligible academic impact.

From an analytics workflow perspective, SQL proved highly effective for cohort segmentation and raw data pre-processing, while Tableau successfully translated these queries into intuitive, interactive visual stories. This combined approach equips administrators with clear points of intervention to improve student outcomes.